The Impact of Meditation App Icons on Likelihood of Use

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Introduction

Meditation apps have become crucial in digitally expanding access to mental health and mediation practices–with Headspace and Calm accounting for 96% of daily active mental health app users, according to a 2020 research study (Wasil AR (2020 Nov), Figueroa CA (2020 Jun)). As a Headspace user, I frequently evaluate why I do not use the headspace app as often as I would like, given that its user interface is accessible and engaging with an overall welcoming aesthetic theme. I gathered that a good proportion of my peers and professors considered using or used the Calm app, and had vaguely heard of Headspace. As there are upwards of 27 apps within the top searches on Apple App Store and Google Play Store, I wondered what set the Calm app apart from Headspace, the app that I use, or another alternative, such as the Healthy Mind app.

This study analyzes the impact of the appearance of meditation app icons on individuals' likeliness of using the app, by comparing individuals' likeliness of use ratings, which range ordinally between 1, very unlikely, and 5, very likely, for three meditation apps–Calm, Headspace, and Healthy Mind. We specifically ask whether mediation app appearance has any effect at all on individuals' likelihood of using the app by choosing three apps with different aesthetic qualities. The Calm app features a light-colored background with white cursive text, the Headspace app features a white background with an orange smiling face, and the Healthy Mind app features a dark-colored background with two white printed letters. As we cannot draw any causative conclusions about which specific aesthetic qualities (such as light/dark color, cursive/print text, or text/no text) directly impact likeliness of use, we evaluate whether the appearance of an app has any impact at all. The null hypothesis of our study is that the average likelihood of use value will be the same between the three meditation app icon appearances. Our research hypothesis is that the meditation app appearance does have an impact on individuals' likeliness of using it, rather than there being no relationship at all, that at least one mean likelihood of use value will be different from the others.

Method

In order to evaluate our research hypothesis, we conduct a randomized one-way experiment with the Calm, Headspace, and Healthy Mind app icons as the levels of our app appearance factor, and with likelihood of use—measured between 1 (very unlikely) and 5 (very likely)—as our response. The experiment was conducted using an anonymous online survey using Qualtrics. Individuals taking the survey first gave their consent in participation, and were asked whether they were above the age of 18. The survey's data collection portion consisted of two parts: (1) a picture of one of the three meditation apps as stimuli, followed by (2) the likert-scale question that asked respondents to indicate how likely they were to use the app they were shown. The individuals who took the survey served as the experimental units and were randomly assigned to see one of the meditation app icons, subsequently answering their likelihood to use the app they were shown. We collected 30 responses, 11 of whom were shown the Calm app icon as stimuli, 10 of whom were shown the Headspace app, and 9 of whom were shown the Healthy Mind app, and every respondent then answered the question "How likely are you to use this app to practice meditation?" by indicating a 1-5 rating.

Results

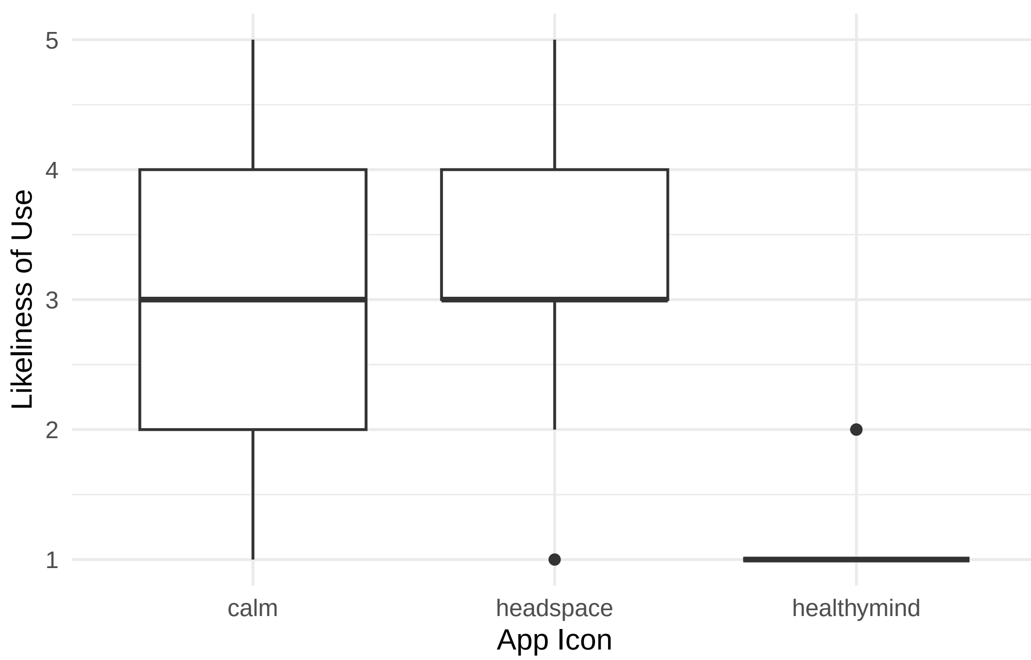


Figure 1: Boxplots

The boxplots in Figure 1 visualizes the distribution of individuals' responses depending on the app icon that the individual received as stimuli. This exploratory visualization demonstrates that the median likelihood of use values for the Calm app and the Headspace app were the same, while the median likelihood of use value for the Healthy Mind app was comparatively lower. The spreads for the likelihood of use values are different for each app, with the Calm app and Headspace app indicating response values distributed between 1 and 5, while Healthy Mind indicated response values strictly between 1 and 2. From this boxplot visualization alone, it looks as though meditation app icon appearance does have an impact on individuals' likelihood of using the app, with Healthy Mind being the app that individuals are less likely to use, compared to Calm and Headspace. To assess the statistical significance of the differences indicated in our exploratory visualization, we calculate an F statistic from our data, using an Analysis of Variance (ANOVA) test.

App	n	mean	sd
calm	11	2.909091	1.3003496
headspace	10	3.300000	1.2516656
healthymind	9	1.222222	0.4409586

Figure 2: Descriptive Statistics

We begin by evaluating whether our ANOVA assumptions are met. We find that the N condition for similar variances is generally met, as our Q-Q plot demonstrates that the distribution of our sample residuals approximately follow a normal distribution. Our residuals are indeed left-skewed, however, as the negative residuals of our model are more negative than the positive residuals are positive. This left-skewness in the distribution of the residuals is explicit when plotting our residuals in a histogram, which also allows us to assess and confirm the Z condition for zero mean residuals. The S condition for similar variances is not quite met, as the standard deviation for the Calm and Headspace apps are both greater than two times the standard deviations for the Healthy Mind responses. As our sample size of 30 is not particularly large, we will proceed with the ANOVA with the understanding that the condition for similar variances was not met. Otherwise, we assume constant effects, additive effects, and that the residuals are independent—that no one residual is dependent on another—with the consideration that the sampling technique used to collect the data was a convenience sampling with the students in SDS290.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
factor(App)	2	22.90202	11.451010	9.494262	0.0007545
Residuals	27	32.56465	1.206098	NA	NA

Figure 3: ANOVA Table

Upon conducting an ANOVA test, we found that meditation app icon appearance does have a statistically significant effect on individuals' likelihood of using the app—that at least one of the mean likelihood of use values was different from the other two, depending on meditation app icon appearance F(27, 2) = 9.494262, p = 0.0007545. We use Fisher's Least Significant Difference (Fisher's LSD) to control for the familywise error rate from our multiple significance tests, and to determine exactly which pairwise comparisons can be considered significantly different. The Fisher's LSD for each app pair, Calm/Headspace, Calm/Healthy Mind, and Headspace/Healthy Mind, is shown in Figure 4, listed next to the pair's difference in mean response values. Fisher's LSD reasons that a difference in a pairwise comparison is significant as long as it is larger than the pair's associated margin of error. We thus find that there is a statistically significant difference between the average likelihood of use between the Calm app and the Healthy Mind app, and between the Headspace app and the Healthy Mind app, but no statistically significant difference between the average likelihood of use between the Calm app and the Headspace app.

Comparison	LSD	Differences
Calm/Headspace	0.9845697	0.390909
Calm/HealthyMind	1.0128160	1.686869
Headspace/HealthyMind	1.0353536	2.077778

Figure 4: Fisher's LSD Table

Comparison	LowerBound	UpperBound	EffectSize
Calm/Headspace	-0.5936607	1.375479	0.3559459
Calm/HealthyMind	0.6740530	2.699685	1.5359946
Headspace/HealthyMind	1.0424244	3.113132	1.8919405

We are 95% confident that the true mean difference in likelihood of use between the Calm app and the Headspace app is between -0.594 and 1.375 units on the likelihood of use scale. Considering that this confidence interval includes 0, there is no evidence that there is a statistically significant difference in the likelihood of use between the Calm and Headspace apps due to app icon appearance. Between Calm and Healthy Mind, we are 95% confident that the likelihood of use difference is between 0.674 and 2.700 units on the survey question scale, and we are 95% confident that the difference in likelihood of use is between 1.042 and 3.113 for Headspace and Healthy Mind.

The difference between the likelihood of use of the Calm app and Headspace app was 0.356 times the size of the typical within group deviations in likelihood of use. Between Calm and Healthy Mind, the differences in their likelihood of use was 1.536 times the within group deviations in likelihood of use. And lastly, between Headspace and Healthy Mind, the difference in likelihood of use was 1.892 times the size of the typical within group deviations in likelihood of use.

Conclusion

This study conducted an analysis of whether different meditation app icon appearances resulted in different likelihoods of use by individuals, measured between 1, very unlikely, and 5, very likely. While the Headspace app and Calm app did not have statistically significant differences in likelihood of use ratings, the Healthy Mind had a significantly different likelihood of use rating from both the Headspace app and Calm app. Given that the treatment condition of which meditation app icon was shown to whom was randomized, this indicates that there is a causative relationship between mediation app icon appearance and individuals' likelihood of using the app.

Although we do find statistically significant differences between the likelihood of use between the different meditation apps, we cannot clearly identify which features of the meditation app icon resulted in these different ratings. One way to improve upon our study would be to implement a two-way factorial design to examine the interaction between specific qualities of the meditation app icons, such as color lightness/darkness, or text/lack of text on the icon. This way, we would be able to gauge which features are more desirable and directly impact the likelihood of use ratings. Another limitation to our study is our sampling technique. For this study, 30 observations were taken, with the 30 observations being from individuals from one statistics class at Smith College. In order to draw conclusions more accurately to true population opinions, one would improve upon this study by surveying a sample size greater than 30, and of individuals of different genders and ages.

R appendix

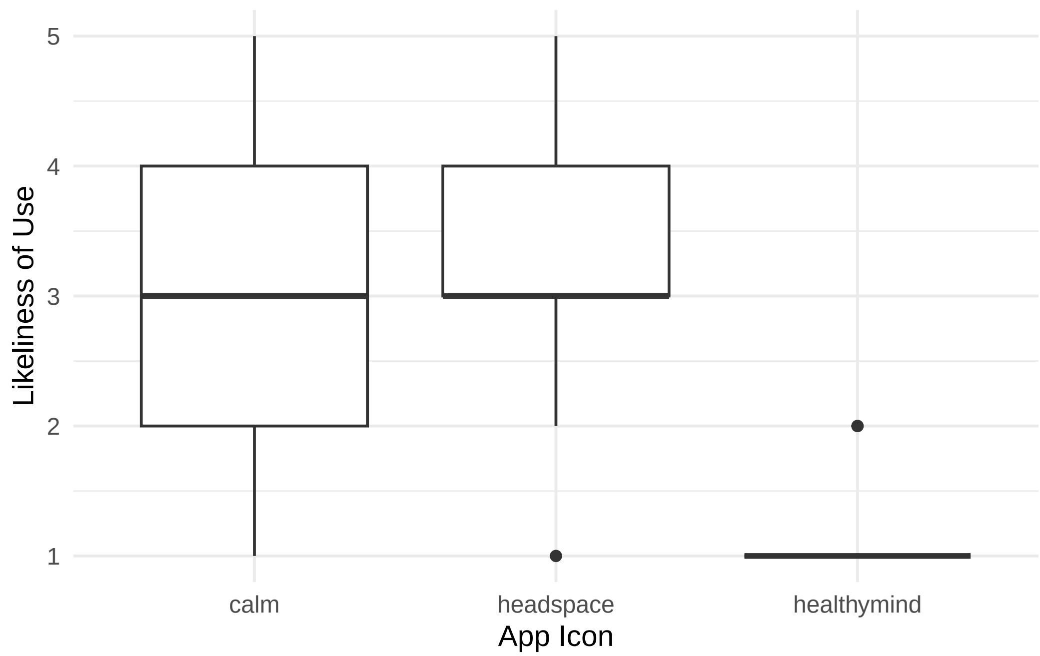
```r
#Necessary packages
library(readr)
library(car)
library(readr)
library(ggplot2)
library(tidyverse)
library(knitr)
library(kableExtra)
library(Stat2Data)
library(broom)
library(dplyr)

#reading in data
data <- read_csv("meditation.csv")
```

```
Rows: 30 Columns: 2
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (1): App
dbl (1): Likeliness

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
ggplot(data, aes(x = factor(App), y = Likeliness)) +
  geom_boxplot() +
  theme_minimal() +
  labs(x = "App Icon",
       y = "Likeliness of Use")
```

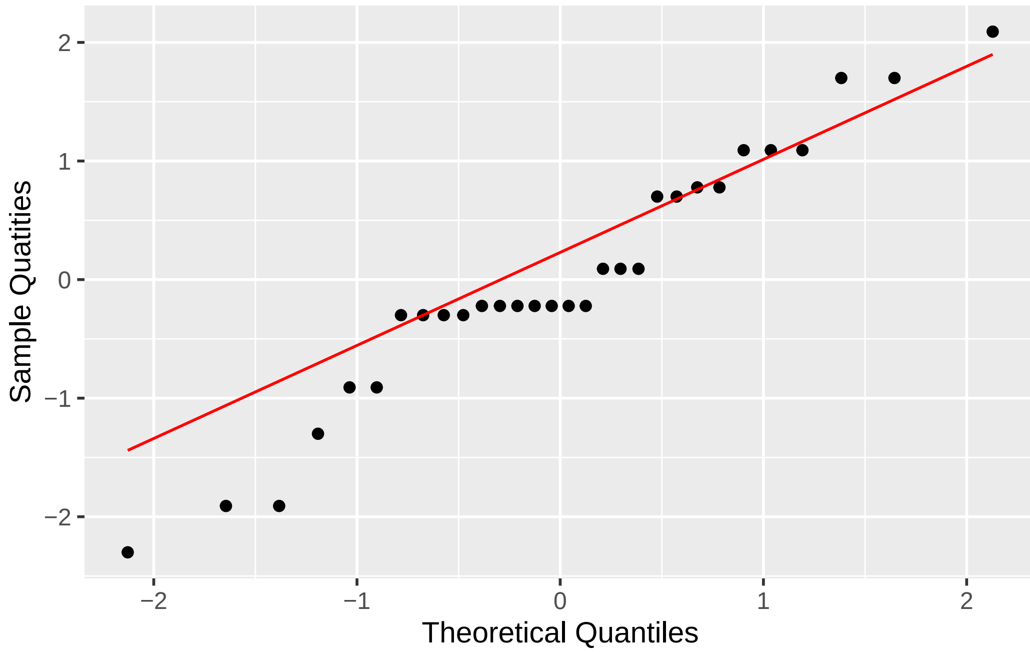



```
summarytable <- data %>%
  group_by(App) %>%
  summarise(n = n(),
            mean = mean(Likeliness),
            sd = sd(Likeliness))
kable(summarytable)
```

App	n	mean	sd
calm	11	2.909091	1.3003496
headspace	10	3.300000	1.2516656
healthymind	9	1.222222	0.4409586

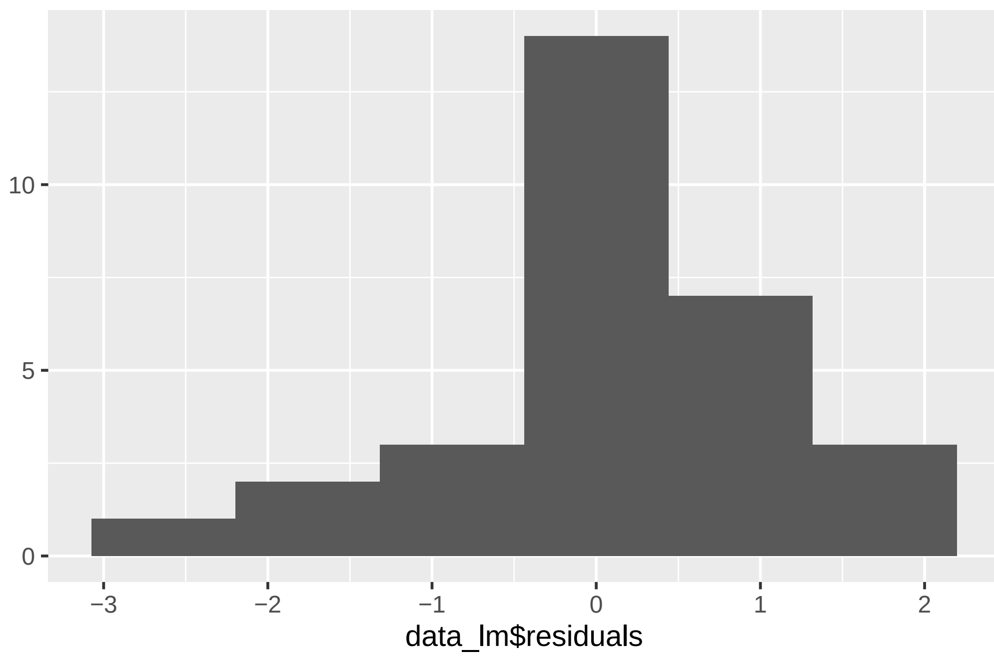
```
#fitting the model
data_lm <- lm(Likeliness ~ factor(App), data = data)

#normality check qq plot of residuals
data_lm |>augment() |>ggplot(aes(sample = .resid)) + geom_qq() +geom_qq_line(col= "red") + xl
```




```
#zero mean residuals check histogram of residuals
qplot(x = data_lm$residuals, bins = 6)
```

```
Warning: `qplot()` was deprecated in ggplot2 3.4.0.
```




```
#anova test
anova <- anova(data_lm)

kable(anova)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
factor(App)	2	22.90202	11.451010	9.494262	0.0007545
Residuals	27	32.56465	1.206098	NA	NA

```
# Calculating Fisher's LSD
mean_calm <- 2.909091
mean_headspace <- 3.300000
mean_healthymind <- 1.222222

MSE <- 1.2061
df_E <- 27
t <- qt(.975, df_E)

ncalm <- 11
nhead <- 10
nhealth <- 9
```

```r
LSDcalmhead <- t*sqrt(MSE)*sqrt((1/ncalm)+(1/nhead))
LSDcalmhealth <- t*sqrt(MSE)*sqrt((1/ncalm)+(1/nhealth))
LSDhealthhead <- t*sqrt(MSE)*sqrt((1/nhealth)+(1/nhead))
LSD <- c(LSDcalmhead, LSDcalmhealth, LSDhealthhead)

# mean differences in likelihood of use due to app icon appearance
# LSD is 1.300986
diffcalmhead <- mean_headspace - mean_calm
diffcalmhealth <- mean_calm - mean_healthymind
diffheadhealth <- mean_headspace - mean_healthymind
Differences <- c(diffcalmhead, diffcalmhealth, diffheadhealth)
Comparison <- c("Calm/Headspace", "Calm/HealthyMind", "Headspace/HealthyMind")
df = data.frame(Comparison, LSD, Differences)

kable(df)
```

Comparison	LSD	Differences
Calm/Headspace	0.9845697	0.390909
Calm/HealthyMind	1.0128160	1.686869
Headspace/HealthyMind	1.0353536	2.077778

```r
#confidence interval
ci_calmhead_lb <- diffcalmhead - LSDcalmhead
ci_calmhead_ub <- diffcalmhead + LSDcalmhead
ci_calmhealth_lb <- diffcalmhealth - LSDcalmhealth
ci_calmhealth_ub <- diffcalmhealth + LSDcalmhealth
ci_headhealth_lb <- diffheadhealth - LSDhealthhead
ci_headhealth_ub <- diffheadhealth + LSDhealthhead
LowerBound <- c(ci_calmhead_lb, ci_calmhealth_lb, ci_headhealth_lb)
UpperBound <- c(ci_calmhead_ub, ci_calmhealth_ub, ci_headhealth_ub)
#effect size
Dcalmhead <- diffcalmhead/sqrt(MSE)
Dcalmhealth <- diffcalmhealth/sqrt(MSE)
Dheadhealth <- diffheadhealth/sqrt(MSE)
EffectSize <- c(Dcalmhead, Dcalmhealth, Dheadhealth)
df2 = data.frame(Comparison, LowerBound, UpperBound, EffectSize)
kable(df2)
```

Comparison	LowerBound	UpperBound	EffectSize
Calm/Headspace	-0.5936607	1.375479	0.3559459
Calm/HealthyMind	0.6740530	2.699685	1.5359946
Headspace/HealthyMind	1.0424244	3.113132	1.8919405

Figueroa CA, Aguilera A. 2020 Jun. "The Need for a Mental Health Technology Revolution in the COVID-19 Pandemic." *Front Psychiatry.*, 2020 Jun. https://doi.org/10.3389/fpsyt.2020.00523.

Wasil AR, Patel R, Gillespie S. 2020 Nov. "Reassessing Evidence-Based Content in Popular Smartphone Apps for Depression and Anxiety: Developing and Applying User-Adjusted Analyses." *J Consult Clin Psychol.*, 2020 Nov. https://doi.org/10.1037/ccp0000604.2020-65097-001.